

## Prognostic Factors in Distant Metastasis

The presence of distant metastases portends the worst prognosis, with mean survival rates measured in months rather than years. Site of metastasis is the most important prognostic factor in the AJCC melanoma staging system; visceral metastases are associated with a poorer prognosis compared to non-visceral sites (such as skin, subcutaneous tissue, and distant lymph nodes). Other significant prognostic variables include the number of metastatic sites and surgical resectability. Solitary metastases that are resected after radiologic confirmation of stability over 3 to 6 months have been associated with prolonged survival in some patients, although there is no strong evidence to date that asymptomatic detection improves overall survival. Elevated serum lactate dehydrogenase (LDH) levels are also associated with a worse prognosis.

## Clinical Course

### *Stages I and II Melanoma*

The prognosis for patients with thin primary melanoma ( $< 1$  mm in Breslow depth) and localized disease is excellent, with long-term survival achieved in more than 90% of cases. However, approximately 15% to 36% of patients with stage I or II melanoma will experience local, regional, or distant recurrence during their clinical course. Melanoma typically recurs in a predictable pattern—first locally and regionally, then at distant sites. Notably, melanoma can also bypass regional lymph nodes through direct hematogenous dissemination.

Most recurrences occur within the first 1 to 5 years after diagnosis and treatment, depending on tumor thickness and other prognostic features of the primary lesion. Nonetheless, late recurrences (10 or more years after initial diagnosis) occur in approximately 1% to 5% of cases. The majority of tumors associated with late recurrence are of intermediate thickness (mean 1.6 mm, typically ranging from 0.70 to 3.0 mm), with a mean time to recurrence of 14.3

years. "Ultra-late" recurrences occurring more than 15 years after diagnosis are extremely rare but have been documented.

### ***Stage III Melanoma***

Data from the Sunbelt Melanoma Trial involving over 1,100 patients show that recurrence rates are significantly higher in early stage III melanoma (micrometastatic nodal disease detected by sentinel lymph node biopsy) compared to SLN-negative melanoma (15.5% vs. 6.0%, respectively). Patients with SLN-positive disease are also more likely to develop distant metastases rather than locoregional recurrence alone (67% vs. 46%). Pooled analyses estimate 5-year overall survival rates of 93% to 88% for SLN-negative patients, compared to 51% to 67% for those with a positive SLN.

Stage III melanoma represents a heterogeneous group of patients with highly variable clinical outcomes. This diversity is reflected in 5-year survival rates, which range from 70% for patients with microscopic nodal involvement to significantly lower rates in those with more extensive nodal or in-transit disease.